

August 29, 2024 04:06 AM GMT

NVIDIA Corp. | North America

NVIDIA strong quarter sets us up for Blackwell ramp

WHAT'S CHANGED

NVIDIA Corp. (NVDA.O)	From	To
Price Target	\$144.00	\$150.00

NVIDIA revenue was again \$2 bn above guidance, and guidance for next quarter was for \$2.5 bn growth (vs pattern of \$2 bn). Gross margin pressure in 2h seems transitional in nature, and we are set up for a strong CY25.

Key Takeaways

- July quarter revenue \$2 bn above guidance, again; October guidance for \$2.5 bn of revenue growth, higher than prior quarters
- Gross margin pressure as guided, but a little more persistent in 2h; we move very slightly lower for CY25
- Stock reacting negatively to a good quarter can feed cautious sentiment, but Blackwell is around the corner and should be a positive through all of next year
- Remain OW, remains our Top Pick in semis

Expectations become more challenging as the superlative becomes mundane, but this was still a very strong quarter given the transitional nature of the current environment. With \$2 bn of revenue upside vs. guidance, and guidance for \$2.5 bn of growth, we would have thought the stock would be OK, even with some gross margin headwinds, but we had highlighted coming in some sell-side numbers as high as \$34 bn for October - though that number may still well be achieved.

Numbers come up by less than in the prior quarters, mostly because of gross margin pressures, but the underlying picture here is that this company has been impacted by multiple obstacles in recent quarters - China restrictions that impacted 25% of revenues, shortages of data center space and power, dependence on a Hopper product that will clearly be superseded by Blackwell, and then delays in that Blackwell platform - and none of that has created anything that could remotely be considered a pause in the business in the last 18 months.

Gross margin pressure is a factor that we have talked about for a while, including in our preview, but we are bringing down CY25 GMs slightly to 74%.

When the company introduced Blackwell and made public statements about pricing to good value, it seemed clear that GMs could come down a little, and the China H2O products are also a headwind to margins. The company indicated modest pressure in October and January; vague guidance for gross margins in the mid 70s for the full year, taken literally could mean 72%, but the company was specific that they do not expect it to be that low. Several factors could be a drag on GMs: the

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NVIDIA Corp. (NVDA.O, NVDA US)

Top Pick

Semiconductors | United States of America

Stock Rating	Overweight
Industry View	Attractive
Price target	\$150.00
Shr price, close (Aug 28, 2024)	\$125.61
Mkt cap, curr (mm)	\$3,155,445
52-Week Range	\$140.76-39.23

Fiscal Year Ending	01/24	01/25e	01/26e	01/27e
EPS (\$) **	1.30	2.80	3.78	4.13
Prior EPS (\$) **	-	2.64	3.53	3.70
P/E	51.0	46.9	35.2	32.3
EPS (\$) §	-	2.75	3.81	4.46
Div yld (%)	0.0	0.0	0.0	0.0

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework

** = Based on consensus methodology

§ = Consensus data is provided by Refinitiv Estimates

e = Morgan Stanley Research estimates

QUARTERLY EPS (\$)

Quarter	2024	2025e Prior	2025e Current	2026e Prior	2026e Current
Q1	0.11	-	0.61a	0.80	0.86
Q2	0.27	-	0.68a	0.90	0.96
Q3	0.40	0.68	0.73	0.91	0.97
Q4	0.52	0.72	0.78	0.91	0.98

e = Morgan Stanley Research estimates, a = Actual Company reported data

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annual cadence of new products which never allows for the mature H100 yields, the return to a China market with US government enforced value limits, new microprocessors and switches for GB200, are all small factors. Some of these gross margin factors will persist into next year, in our view.

Color on the Blackwell delay was entirely positive, in our view. The company said that the mask change was not a functionality issue, but a change to enable yield improvements, which is consistent with what we had heard, and the "several billion in revenue" for Blackwell in the January quarter is at the higher end of expectations. For the company to be that specific at this point, confidence must be high, and clearly demand for both Hopper and Blackwell remains strong.

There were several other important growth metrics on the call:

- Sovereign business this year will be "low double digit" billions in revenue this year (was high single digits last quarter)
- Inference for the trailing 12 months is above 40% of revenues (was 40% in 1h, implying slight increase as a percentage of revenue, which in our view should help to answer the ROIC questions given that inference is directly tied to usage)
- Ethernet for ai applications, including spectrum X, more than doubled q/q
- The software business will exit the year at a \$2 bn run rate

Details on the quarter: July revenue of \$30.04bn (up 15.3% q/q and 122.4% y/y) beat Street expectations of \$28.723bn and our estimate of \$28.171bn. By segment, Data Center revenue of \$26.272bn was up 16.4% q/q and 154.5% y/y, Gaming was up 8.8% q/q and 15.8% y/y to \$2.880bn, Professional Visualization was up 6.3% q/q and up 21.1% y/y to \$454mn, Automotive was up 5.2% q/q and 36.8% y/y to \$346mn, and OEM & Other was up 12.8% q/q and up 33.3% y/y to \$88mn. Gross margin of 75.7% was in line with the Street and our estimate of 75.7% and 75.6% respectively. Non-GAAP EPS of \$0.68 beat both the Street and our estimate of \$0.65 and \$0.63, respectively.

NVIDIA guided to a a larger \$2.5bn sequential revenue increase; to \$32.5bn at the midpoint for July up 8.2% q/q and 79.4% y/y) came in ahead of the Street and our estimate of \$31.640bn and \$30.221bn, respectively, with data center once again the biggest contributor to growth. The company guided gross margin to 75.0%, compared to the Street at 75.5% and our estimate of 75.7%.

Raising estimates: We are increasing our growth estimates for CY25, but continue to remain conservative beyond the shorter term forecast window. We are increasing estimates for January quarter revenue from \$32,244bn to \$35.382bn and non-gaap EPS from \$0.72 to \$0.78. For FY25 our revenue and non-gaap EPS move from \$116.679/\$2.64 to \$123.999/\$2.80, and FY26 comes up to \$166.926bn/\$3.78 from \$152.649bn/\$3.53 previously.

Thoughts on the stock: We were surprised at the after hours sell off; while we had seen previews calling for \$34 bn in October, our sense is that most expectations were somewhat lower, and we would have thought that these numbers would be good enough.

As numbers get bigger, the growth percentages have to slow, but any second derivative based thought process has to take into account that the company announced Blackwell in March, offering substantially better value than Hopper, and yet growth in Hopper continues to be exponential - to the point that sequential growth forecasts are accelerating in absolute dollars in a quarter that has zero Blackwell revenue. Anticipation of Blackwell remains very high.

We think that Blackwell will be an accelerant to growth next year, and we expect NVIDIA to gain share slightly in processors, vs. custom ASIC or alternative merchant solutions

In terms of our price target, we are reducing our MW multiple expectations slightly from 43x to 42x (which is 40x non GAAP, in line with trailing 5 year average vs prior PT of 40.8x a slight premium). As uncertainties around the medium term gross margin profile somewhat limit willingness to pay the same premium for incremental revenues. That brings the target from \$144 to \$150.

Risk Reward – NVIDIA Corp. (NVDA.O) Top Pick

OW as large language model enthusiasm is transforming cloud capex

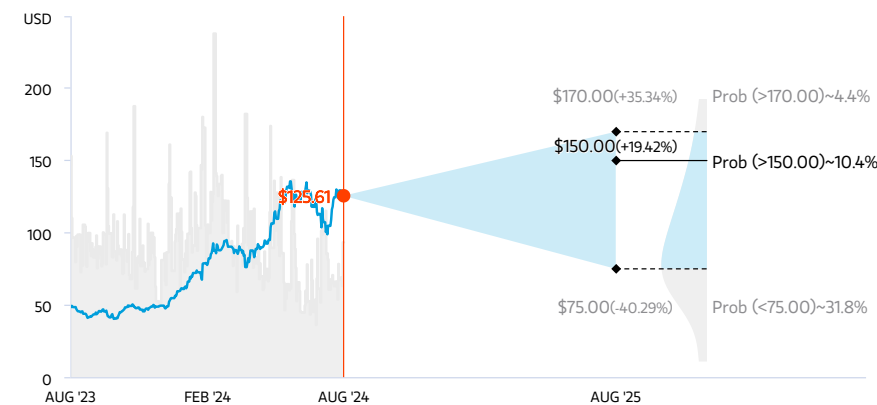
PRICE TARGET \$150.00

~42x our MW CY25 EPS estimate of \$3.57, a premium to peers, reflecting the expansion in all AI names as well as our higher conviction in estimates given NVIDIA's higher AI exposure. We believe that NVIDIA should trade at a premium given its higher probability of upward revisions in the near term.

Consensus Price Target Distribution



RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 28 Aug 2024. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 28 Aug 2024. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

BULL CASE \$170.00

~40x bull case MW CY25 EPS of ~\$4.24

Bull case has DC revenues continuing to grow through 2025. Upside from networking, GB200 based systems, and software create potential for a full stack AI computing company worthy of an even greater valuation premium

- Higher margin data center and AI-focused software and services growth accelerates
- GPU based AI PC gains traction, widely increasing the client TAM
- Automotive opportunity takes off, allowing the company to earn recurring, per-car licensing revenue

BASE CASE \$150.00

~42x our MW CY25 EPS of \$3.57

43x valuation is a premium to peers, reflecting the expansion in all AI names as well as our higher conviction in estimates given NVIDIA's higher AI exposure. We believe that NVIDIA should trade at a premium given its higher probability of upward revisions in the near term.

- Revenue grows by 103.5% in 2024 and 34.6% in 2025
- Datacenter continues to grow significantly through 2024 and into the first half of 2025 as supply remains constrained
- NVDA maintains leadership in training workloads

\$150.00

BEAR CASE

\$75.00

~28x bear case MW CY25 EPS of ~\$2.72

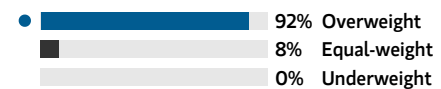
Two key debates both go the wrong direction, causing investors to question future prospects for growth

- Growth in DC slows substantially as supply catches up to demand faster than anticipated
- AI development costs come down materially, a strong competitor enters the market to take market share, or customers begin insourcing custom hardware solutions
- Greater than expected impact from export controls

OVERWEIGHT THESIS

- We expect NVDA's Data Center business to drive much of the growth over the next 5 years, as enthusiasm for generative AI has created a strong environment for AI/ML hardware solutions - NVDA's being one of the most important. Incremental opportunities in AI/ML software & services, networking, and ADAS can drive growth even higher.
- Blackwell availability later this year should further cement NVDA's competitive position, with higher dollar content through a greater mix of system sales

Consensus Rating Distribution



MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era: *Positive*
Pricing Power: *Positive*
Secular Growth: *Positive*

View descriptions of Risk Rewards Themes [here](#)

Risk Reward – NVIDIA Corp. (NVDA.O)

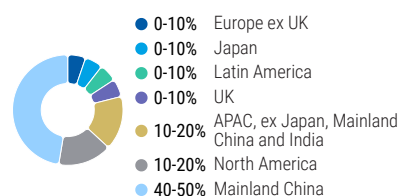
KEY EARNINGS INPUTS

Drivers	2024	2025e	2026e	2027e
GAAP Revenue (\$, mm)	60,922	123,999	166,926	179,981
MW Gross Margin (%)	73.6	75.4	73.6	73.6
MW EPS (\$)	1.21	2.68	3.57	3.89
Inventory (\$, mm)	5,282	8,558	10,290	11,749
DOI	114.4	99.5	83.1	88.0

INVESTMENT DRIVERS

- Growth in AI capex from customers
- Next gen GPUs continue to outpace the competition
- Systems approach allows for higher monetization over time
- New drivers emerge for Nvidia such as AI PCs, autonomous vehicles, robotics, and software

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

5/5 BEST	24 Month Horizon	4/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Growth in training and inference propel data center revenue
- Gaming sales accelerate as GPU based AI PCs gain traction
- DGX platforms gain traction, especially among enterprise customers

RISKS TO DOWNSIDE

- AI end markets don't materialize as expected, customers sharply reduce GPU purchases
- AMD reemerges as a viable GPU competitor
- Cloud customers are able to develop competitive custom hardware

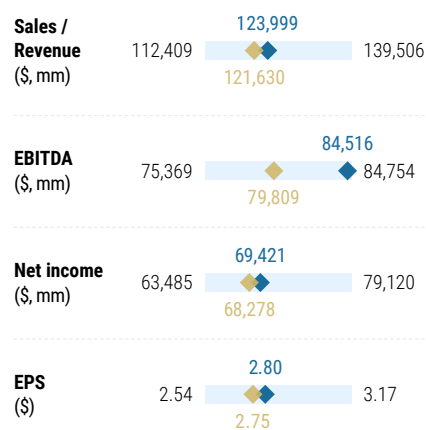
OWNERSHIP POSITIONING

Inst. Owners, % Active	52.4%	
HF Sector Long/Short Ratio	2.1x	
HF Sector Net Exposure	25.3%	

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Jan 2025e



Source: Refinitiv, Morgan Stanley Research

Financial Summary

Exhibit 1: Income Statement Summary

NVDA: Snapshot for the quarter ended July 2024

Income Statement									
Qtr Results:	Actual			Last Qtr	QoQ	Last Yr	YoY	Consensus	MSe
Revenue	\$28,171			\$26,044	8.2%	\$13,507	108.6%	\$0	\$28,171
Gross Margin	75.6%			78.9%	-334	71.2%	443	0.0%	75.6%
EPS	\$0.63			\$0.61	\$0.01	\$0.27	\$0.36	\$0.65	\$0.63
Next Q Guide:	Low	High	Midpoint						
Revenue	\$29,616	\$30,825	\$30,221		7.3%	\$18,120	66.8%	\$31,640	\$30,221
Gross Margin	66.5%	67.5%	75.7%		8	75.0%	72	75.5%	75.7%
EPS						\$0.40		\$0.71	\$0.68

Source: Company data, Morgan Stanley Research estimates

Exhibit 2: Income Statement Variance

MW Financials, \$ in Millions except per share

	Reported FY2Q25A	MS Est. FY2Q25A	Difference Act vs. Est.	Per share Difference
Total Revenue	30,040	28,171	1,869	0.08
Cost of Sales	7,466	7,027	439	0.02
Gross Profit	22,691	21,262	1,429	0.06
Gross Margin	75.5%	75.5%	0.1%	
R&D	3,073	3,188	-115	NM
SG&A	833	812	21	NM
Other Opex	0	0	0	NM
Total Op Expenses	3,906	4,000	-94	NM
Operating Income	18,785	17,262	1,523	0.06
Operating Margin	62.5%	61.3%	1.3%	
Interest income	444	314	130	NM
Interest expense	-61	-66	5	NM
Other expense (Income)	189	53	136	NM
Income Before Taxes	19,168	17,510	1,658	0.07
Tax Rate	13.6%	17.0%	-3.4%	
Income tax provision	2,609	2,977	-368	-0.01
Net Income	16,559	14,534	2,026	0.08
MW EPS	\$0.67	\$0.58	\$0.08	\$0.08

Source: Company data, Morgan Stanley Research estimates

Projected Financial Statements

Exhibit 3: NVDA: Projected Income Statement

Nvidia (NVDA-US)																													
Nvidia Corporation Income Statement		2023:1F	2023:2F	2023:3F	2023:4F	2024:1F	2024:2F	2024:3F	2024:4F	2025:1F	2025:2F	2025:3F	2025:4F	FY2022	2023FY	2024FY	2025FY	2026FY	2027FY										
		Apr/23	Jul/23	Oct/23	Jan/24	Apr/24	Jul/24	Oct/24	Jan/25	Apr/25	Jul/25	Oct/25	Jan/26	2022	2023	2024	2025	2026	2027										
		1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26	4Q26	2022	2023	2024	2025	2026	2027										
(\$ in millions; FY ends January)																													
Income Statement - GAAP																													
Revenue		7,192	13,507	18,120	22,101	26,044	30,040	32,534	35,382	38,574	42,556	42,810	43,008	26,874	60,302	123,999	166,926	179,981	208,814										
Y/Y Change		-13.2%	101.8%	205.8%	268.3%	262.1%	122.4%	47.8%	60.1%	48.1%	41.6%	31.6%	21.5%	0.2%	125.8%	103.8%	34.6%	7.8%	15.6%										
G/O Change		18.9%	67.9%	34.2%	22.2%	17.6%	15.3%	8.3%	8.9%	9.7%	10.3%	6.9%	8.5%	11.61%	16.61%	10.37%	44.59%	48.07%	55.66%										
COGS		2,544	4,045	4,719	5,312	5,638	7,068	7,807	9,573	10,327	11,331	11,492	11,510	11,618	16,261	30,973	44,599	46,075	55,664										
Gross Profit		4,648	9,462	13,400	16,789	20,406	22,972	24,727	25,809	28,247	31,225	31,318	31,498	15,262	44,041	83,000	122,327	133,906	153,150										
Gross Margin		64.6%	70.1%	74.2%	76.0%	78.4%	76.5%	75.1%	74.5%	72.9%	73.2%	73.4%	73.2%	56.8%	72.7%	70.1%	73.3%	73.3%	73.4%										
R&D as % of Revenue		1.87%	2.04%	2.23%	2.45%	2.72%	3.00%	3.00%	3.10%	3.31%	3.52%	3.52%	3.52%	7.33%	6.81%	6.32%	6.24%	6.24%	6.24%										
SG&A as % of Revenue		0.63%	0.62%	0.69%	0.71%	0.77%	0.82%	0.80%	0.80%	0.80%	0.80%	0.80%	0.80%	1.40%	1.35%	1.29%	1.30%	1.30%	1.30%										
Other Opex as % of Revenue		8.6%	4.6%	3.8%	3.2%	3.0%	2.9%	2.6%	2.3%	2.1%	2.0%	2.1%	2.3%	2.53%	2.05%	1.96%	1.96%	1.96%	1.96%										
Other Opex		2,508	2,662	2,983	3,176	3,797	3,992	4,230	4,536	4,739	4,905	5,000	5,195	11,153	11,629	12,609	13,010	12,440	15,165										
Operating Income		2,140	6,800	10,417	13,615	16,908	18,642	19,907	21,279	23,817	26,340	26,368	26,361	4,225	32,973	73,377	102,317	106,416	127,885										
Operating Margin		29.8%	50.3%	57.5%	61.6%	64.6%	62.1%	61.2%	61.0%	61.0%	61.0%	61.0%	61.0%	15.7%	54.1%	59.9%	61.4%	60.0%	61.8%										
Other Operating Income		(15)	59	(86)	260	75	189	103	112	148	126	136	125	(47)	238	480	534	522	520										
Interest Income		150	187	234	294	359	444	317	395	481	583	701	845	267	865	1,515	2,610	3,379	3,379										
Interest Expenses		(66)	(65)	(63)	(63)	(64)	(61)	(66)	(66)	(66)	(66)	(66)	(66)	(60)	(237)	(257)	(264)	(264)	(264)										
Profit Before Tax		2,209	6,961	10,622	14,106	17,279	19,214	20,261	21,791	24,086	26,983	27,129	27,265	4,180	33,818	74,475	105,356	110,933	131,621										
Income Tax		166	793	1,279	1,821	2,368	2,615	3,444	3,860	4,084	4,587	4,612	4,635	1,962	4,599	12,150	17,917	19,079	21,827										
Effective tax rate/GAAP %		7.5%	11.4%	12.2%	12.9%	13.8%	13.6%	17.0%	17.0%	17.0%	17.0%	17.0%	17.0%	45.8%	7.6%	15.7%	17.2%	17.2%	17.2%										
Net Income		2,043	6,168	9,343	12,285	14,911	16,599	16,817	18,028	19,996	22,396	22,517	22,580	2,218	36,799	62,325	87,439	93,834	109,794										
EPS - GAAP		\$0.08	\$0.25	\$0.37	\$0.49	\$0.60	\$0.67	\$0.68	\$0.72	\$0.81	\$0.91	\$0.92	\$0.92	\$0.174	\$1.174	\$2.427	\$3.577	\$3.879	\$4.461										
EPS - GAAP excluding NTD royalty @ 30% tax		\$0.08	\$0.25	\$0.37	\$0.49	\$0.60	\$0.67	\$0.68	\$0.72	\$0.81	\$0.91	\$0.92	\$0.92	\$0.174	\$1.174	\$2.427	\$3.577	\$3.879	\$4.461										
Dividend Paid		(99)	(99)	(99)	(99)	(99)	(99)	(99)	(99)	(99)	(99)	(99)	(99)	(99)	(99)	(99)	(99)	(99)	(99)										
Dividend Per Share		\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.01	\$0.01	\$0.01	\$0.01	\$0.01	\$0.01	\$0.01	\$0.01	\$0.01	\$0.01	\$0.01	\$0.01	\$0.01										

Source: Company data, Morgan Stanley Research estimates

Exhibit 4: NVDA: Projected Balance Sheet

Nvidia (NVDA-US)																																
Nvidia Corporation Balance Sheet																																
Actual Periods (7 ends January)																																
Financial Year	2021 1F	2022 1F	2022 2F	2022 3F	2022 4F	2023 1F	2023 2F	2023 3F	2023 4F	2024 1F	2024 2F	2024 3F	2024 4F	2025 1F	2025 2F	FY2022	2022 1F	2022 2F	2022 3F	2022 4F	2023 1F	2023 2F	2023 3F	2023 4F	2024 1F	2024 2F	2024 3F	2024 4F	2025 1F	2025 2F		
Period	Apr/21	Jul/21	Oct/21	Jan/22	Apr/22	Jul/22	Oct/22	Jan/23	Apr/23	Jul/23	Oct/23	Jan/24	Apr/24	Jul/24	Oct/24	Jan/25	2022	2022	2022	2022	2023	2023	2023	2023	2024	2024	2024	2024	2025	2025		
Assets																																
Cash and cash equivalents	1,288	1,360	3,887	3,513	2,800	5,889	5,079	5,882	5,519	7,289	7,587	8,571	9,389	9,807	10,254	3,889	7,280	26,609	62,695	143,988	213,131											
Accounts receivable	18,192	19,212	16,461	16,424	15,343	19,027	19,241	18,545	17,782	18,924	20,853	21,221				9,807	18,754	26,229	28,262	28,262												
Accounts receivable	1,598	4,650	4,438	5,317	4,089	3,827	4,080	7,086	8,339	9,280	10,365	11,152				5,889	19,645	20,202	20,202	20,202												
Investments	2,225	2,660	2,183	2,660	4,654	1,159																										
Prepaid expenses and other assets	221	361	638	1,175	213	711	872	1,389	1,289	3,088	4,062	4,026	71	310	4,080	4,742	5,764	5,764	5,764	5,764												
Total current assets	38,086	38,282	28,575	27,416	23,223	25,073	24,893	26,797	22,688	44,345	53,729	58,653				23,073	44,345	82,633	145,210	211,996	289,237											
Property, plant and equipment, net	2,509	2,678	2,791	2,791	2,791	1,807											1,807	2,791	2,791	2,791												
Goodwill	4,302	4,349	4,365	4,372	4,372	4,372	4,430	4,430	4,430	4,430	4,430	4,430				4,430	4,430	4,430	4,430	4,430												
Intangible assets	2,434	2,433	2,221	2,220	1,676	1,251	1,650	1,676	1,251	1,650	1,650	1,650				1,650	1,650	1,650	1,650	1,650												
Other assets	5,661	5,882	1,805	4,417	7,209	2,584	9,866	11,134	11,909	11,907	13,899	15,105				8,252	19,207	15,105	15,105	15,105												
Total non-current assets	10,908	14,337	11,275	12,781	16,040	10,187	16,490	16,331	16,490	16,490	16,490	16,490				11,907	28,887	34,476	34,476	34,476												
Total assets	48,994	52,619	39,850	40,197	39,263	35,260	41,383	43,128	39,179	60,835	70,219	75,143				34,980	73,232	117,109	179,686	246,472	318,254											
Liabilities & Shareholders' equity																																
Accounts payable	1,694	1,763	1,959	2,421	1,491	1,193	1,141	1,820	2,359	2,699	2,715	3,660				1,193	2,699	4,718	5,673	6,478	7,475											
Accrued liabilities	1,580	2,052	1,580	1,580	1,580	1,580	1,580	1,580	1,580	1,580	1,580	1,580				1,580	1,580	1,580	1,580	1,580												
Long-term debt	3,812	4,330	5,582	7,373	6,855	5,815	7,260	10,334	9,151	16,831	15,223	13,969				6,855	18,831	15,067	15,067	15,067	17,784											
Other long-term liabilities	10,344	10,842	10,417	10,417	10,417	10,417	10,417	10,417	10,417	10,417	10,417	10,417				10,417	10,417	10,417	10,417	10,417												
Other long-term liabilities	2,278	2,294	2,383	2,352	2,263	2,815	2,676	3,204	3,325	3,661	4,247	4,640				2,815	3,660	4,640	4,640	4,640												
Deferred income taxes																																
Shareholders' equity																																
Common stock																																
Retained earnings																																
Accumulated other comprehensive income																																
Residual earnings																																
Total shareholders' equity																																
Total liabilities and shareholders' equity																																

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options_Probabilities_Exhibit_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR_Themes_Exhibit_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG_Exhibit_Link.pdf](#)
4. View explanation of Theme/Exposure methodology - [ESG_Sustainable_Solutions_External_Link.pdf](#)
5. View explanation of HERS methodology - [ESG_HERS_External_Link.pdf](#)

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(as of July 31, 2024)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1427	38%	335	44%	23%	663	40%
Equal-weight/Hold	1740	46%	347	46%	20%	779	46%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	585	16%	76	10%	13%	233	14%
Total	3,755		758			1676	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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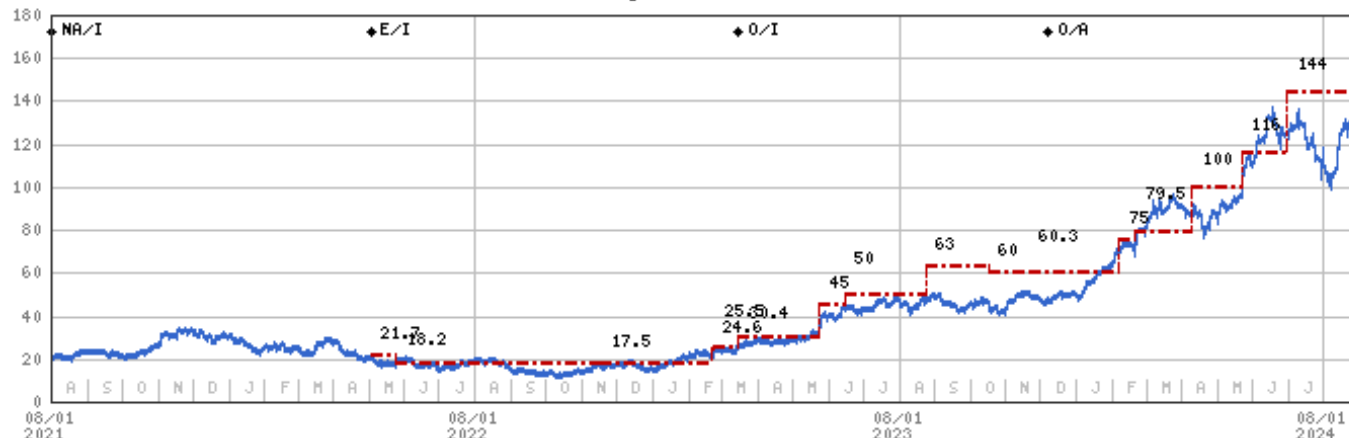
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Stock Price, Price Target and Rating History (See Rating Definitions)

NVIDIA Corp. (NVDA.O) - As of 08/29/24 GMT in USD Industry : Semiconductors



Stock Rating History: 8/1/19 : E/C; 11/25/19 : O/C; 12/18/19 : O/I; 6/16/20 : E/I; 9/13/20 : NA/I; 5/3/22 : E/I; 3/16/23 : O/I; 12/7/23 : O/A

Price Target History: 5/13/19 : 4.25; 8/16/19 : 4.5; 11/15/19 : 5.43; 11/25/19 : 6.47; 2/14/20 : 7.6; 3/18/20 : 6.7; 4/28/20 : 7.05; 5/22/20 : 9.07; 6/16/20 : 9.5; 8/20/20 : 12.68; 9/13/20 : NA; 5/3/22 : 21.7; 5/25/22 : 18.2; 11/17/22 : 17.5; 2/21/23 : 24.6; 2/22/23 : 25.5; 3/16/23 : 30.4; 5/24/23 : 45; 6/15/23 : 50; 8/24/23 : 63; 10/17/23 : 60; 11/21/23 : 60.3; 2/6/24 : 75; 2/21/24 : 79.5; 4/9/24 : 100; 5/22/24 : 116; 6/30/24 : 144

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —
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Joseph Moore		
Advanced Micro Devices (AMD.O)	E (06/09/2024)	\$146.36
Aeva Technologies Inc (AEVA.N)	E (07/19/2021)	\$3.18
Ambarella Inc (AMBA.O)	O (03/29/2016)	\$58.40
Amkor Technology Inc (AMKR.O)	E (11/08/2023)	\$32.63
Analog Devices Inc. (ADI.O)	O (11/16/2023)	\$227.17
Astera Labs Inc (ALAB.O)	E (04/15/2024)	\$43.52
Broadcom Inc. (AVGO.O)	O (06/09/2024)	\$158.18
GlobalFoundries Inc (GFS.O)	O (03/03/2022)	\$45.35
Intel Corporation (INTC.O)	E (02/22/2023)	\$19.61
IonQ Inc (IONQ.N)	E (04/25/2023)	\$7.01
Marvell Technology Group Ltd (MRVL.O)	E (09/14/2015)	\$68.44
Microchip Technology Inc. (MCHP.O)	E (07/10/2024)	\$79.40
Micron Technology Inc. (MU.O)	E (05/20/2024)	\$94.86
Navitas Semiconductor Corp (NVT.S.O)	E (08/28/2023)	\$2.93
NVIDIA Corp. (NVDA.O)	O (03/16/2023)	\$125.61
NXP Semiconductor NV (NXPI.O)	E (04/08/2021)	\$246.91
ON Semiconductor Corp. (ON.O)	U (07/10/2024)	\$74.63
Qorvo Inc (QRVO.O)	O (12/07/2023)	\$112.26
Qualcomm Inc. (QCOM.O)	E (12/07/2023)	\$171.85
Silicon Laboratories Inc. (SLAB.O)	E (01/19/2021)	\$114.11
Skyworks Solutions Inc (SWKS.O)	E (11/28/2018)	\$106.73
Texas Instruments (TXN.O)	U (04/13/2020)	\$207.74
Western Digital (WDC.O)	O (01/23/2020)	\$62.43
Wolfspeed, INC (WOLF.N)	E (12/07/2020)	\$10.34
Lee Simpson		
Arm Holdings plc (ARM.O)	O (07/19/2024)	\$125.32
Cadence Design Systems Inc (CDNS.O)	O (02/14/2024)	\$263.09
Synopsys Inc. (SNPS.O)	O (11/10/2023)	\$509.98

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.

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